

Willow WU

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EDUCATION

Columbia University

M.S. in Computer Science (*Machine Learning Track*)

New York, NY

Sep 2025 – Dec 2026

Hong Kong Baptist University (HKBU)

B.Sc. in Data Science

Hong Kong

Sep 2021 – Jun 2025

EXPERIENCE

Microsoft — Software Engineer Intern

Summer 2026

OfficePLUS AI Assistant

- Contributed to a team refactor of a multi-session, token-streaming LLM chat engine (React/TypeScript, WebSocket RPC, ~2,160-LOC hook), consolidating 7–8 run-tracking tables into a 2-table session-keyed model with invariant-based garbage collection that cannot evict active runs.
- Built a routing-verification probe proving **500/500** concurrent runs routed correctly with **zero** cross-session cross-talk; audited 6 error-fallback branches (2 fired live) to prevent regressions.
- Shipped a change through the full release chain (PR → rebase → Azure DevOps pipeline → signed artifact → handoff); fixed dual-instance concurrency and input-loss bugs.

Columbia University — Teaching Assistant, Introduction to Databases

New York, NY

Prof. Kenneth Ross

Sep 2025 – Present

- Supported **100+** students in relational schema design, SQL implementation, and query-plan analysis; graded exams and debugged indexing/performance issues.

Shenzhen Haohai Zhiyuan — Software Engineer Intern

Shenzhen, China

Nov 2023 – Mar 2024

- Deployed AWS-based IoT solutions and built React Native / Flutter cross-platform interfaces (10+ languages); automated IoT device testing workflows, reducing manual QA.

PROJECTS

FlowTrack — Full-Stack Job-Search Tracker

Columbia ESaaS

2025

- Built a full-stack tracker in **Ruby on Rails** + **PostgreSQL** with a live **Sankey diagram** for Applied → Interview → Offer; **RSpec** + **Cucumber** tests; deployed to Heroku.

Evidence of Life — AI Life-Tracking Web App

Startup Studio

2026

- Built a solo full-stack React/TypeScript SPA on **Supabase** (Auth, Postgres, Edge Functions) with AI-assisted input classification and map/calendar views; **51** passing Vitest tests.

Oracle OS — AI Symbolic-Reflection Product

Hackathon

2026

- Architected a full-stack app (**React 19** + Express/TypeScript) with deterministic engines and zod-validated APIs; isolated LLM calls to reflect/chat endpoints only.

PUBLICATIONS

You Wu, Junwei Su, Sirui Zhang, Zhijian Li. Explainable Dynamic Graph Neural Networks for Predictive Maintenance in Vehicle Chassis Systems. **ACML 2025** (PMLR).

SKILLS

Languages: Python, TypeScript/JavaScript, SQL, Ruby, C++

Web / Full-Stack: React, Node.js, Ruby on Rails, Supabase, Tailwind, shadcn/ui

Testing / Infra: RSpec, Cucumber, Vitest; Azure DevOps, AWS, Heroku, Vercel, Git

Databases: PostgreSQL, SQLite, DuckDB

HONORS & AWARDS

Kaggle Silver Medal (Top 4.2%, 2024) • **MCM Honorable Mention** (Top 20%, 2024) • **CUMCM Provincial 2nd Prize** (Top 16%, 2023) • Merit Scholarship (Top 15%)